Online Learning Impact Analysis

A Project Proposal

Submitted in the partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE WITH SPECIALIZATION IN

BIG DATA ANALYTICS

Submitted by:

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DEPARTMENT OF APEX INSTITUTE OF TECHNOLOGY

PROJECT PROPOSAL

1. Project Title: -

“Online Learning Impact Analysis”

2. Project Scope: -

This project will examine the effects of the **online learning platforms** on the students and determine their patterns of **learning, frequency of usage, and effectiveness using a data-driven analytical methodology.** According to recent Scopus-indexed publications, online learning has been found to increase the availability of access and flexibility, but its effectiveness is vastly different according to the behaviour of the students in engagement, self-regulation, and the quality of interactions (Akpen et al., 2024; De Bruijn-Smolders and Prinsen, 2024).

Similar to the results in Scopus indexed, this project aims to concentrate on the **behavioral dimension of online learning** other than platform-based outcomes. The articles in the Scopus-indexed journals highlight that online learning is a multidimensional behavioral, cognitive, and affective process, and that poor learning habits and distractions adversely affect learning results (Broadbent and Poon, 2015).

Based on this, the self-reported data included in this project with respect to the students in various academic fields is meant to examine the interaction of these students with the online learning platforms. The areas of interest are the length of time devoted to online studies, platform use, the duration of one session, perceived competence, satisfaction rates, and the difficulties associated with the online education. Such demographic attributes as the level of education and the year of study are added with the aim of allowing a comparative analysis.

To fill the gaps found in the literature indexed in Scopus, the project presents a **behaviour-based analytical** viewpoint through the inclusion of the **Learning Fragmentation Index (LFI)** to quantify the interruptions and session discontinuity and the **Replay Dependency Phenomenon (RDP)** to examine dependence on the pause and replay features. This research provides statistical analysis with Python, and visualization with Power BI dashboards to come up with actionable insights to guide evidence-based changes in online learning.

Key Goals:

1. **Planned Data Gathering According to Learning Analytics Research:**
    - Gather student feedback in the form of Google forms with a concentrate on engagement behaviour, consumption patterns, and perceived effectiveness.
    - Cover demographic, behavioral and self-regulated learning indicators that have been noted in the current studies.

2. **Behavioral and Statistical Analysis of Online Learning Usage**:
    - Preprocess survey data with Microsoft Excel or Google Sheets.
    - Apply statistical analysis to find the relationship between frequency of use, learning fragmentation and perceived effectiveness.

3. **Patterns and trends of learning and engagement visualization:**
    - Create interactive dashboards with Power BI to see how learning behavioral patterns and engagement work.
    - Deliver findings presently in a way that they can be used to make academic and institutional decisions, as suggested by analytics-driven research in education.

4. **Learning Challenges and Areas of Improvement.:**
    - Establish behavioral risks including broken learning sessions and replay addiction.
    - Present evidence-based information, which can be in line with the recommendations based on Scopus-indexed articles concerning online learning effectiveness.

3. Requirements: -

- <u>Hardware Requirements</u>
    1. Laptop or Desktop
    2. Internet connection

- <u>Software Requirements</u>
    1. Google Form (for data collection)
    2. Microsoft Excel or Google Sheets (for data cleaning)
    3. Python (for statistical analysis)
    4. Power BI (for dashboard visualization)

STUDENTS DETAILS

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APPROVAL AND AUTHORITY TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Title	Signature (With Date)
Dr. Amardeep Singh	Online Learning Impact Analysis	